SAI ROHITH BHARADWAJ RAMAVARAPU

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EDUCATION

University at Buffalo, SUNY — M.S. Computer Science, Artificial Intelligence *Expected Dec 2026*

VNR Vignana Jyothi Institute of Engineering & Technology — B.Tech, Computer Science *May 2024*

TECHNICAL SKILLS

Languages: Python, C++, SQL, C#

ML & Computer Vision: PyTorch, TensorFlow, OpenCV, Scikit-learn, Hugging Face, CNNs, U-Net, Transformers, Visual SLAM

ML Systems: FastAPI, Docker, MLflow, ONNX, Weights & Biases, NumPy, Pandas, Linux, Git

EXPERIENCE

University at Buffalo — Graduate Research Assistant, Robotics & Vision *Aug 2026 – Present*

- Developing adaptive closed-loop illumination methods to improve Visual SLAM robustness in extreme low-light environments.
- Trained Co-Located Illumination Decomposition models to separate ambient and controllable-light contribution fields from camera observations.
- Implemented behavior-cloning pipelines that learn real-time light-intensity commands from optimized intensity schedules.

UST — Software Engineer, Backend & Analytics *Dec 2024 – Jun 2025*

- Built 12+ REST microservices for carbon-emissions analytics across transportation, waste, and supply-chain workflows.
- Reduced API response latency by approximately 30% through SQL indexing, connection pooling, complexity improvements, and asynchronous data access.
- Implemented JWT authentication, role-based access control, and structured logging for secure, diagnosable services.

Tech Mahindra — AI / NLP Engineer Trainee *Sep 2023 – Nov 2023*

- Built semantic document retrieval for multi-page PDFs using sentence embeddings, chunking, cosine similarity, and vector indexing.
- Developed NLP preprocessing and intent-classification pipelines; recorded an F1 score of 0.89 on the project evaluation set.

South Matrix Institute of Technology — Machine Learning Intern *Jun 2024 – Dec 2024*

- Analyzed 500K+ transaction records with Python and Pandas and engineered behavioral features for clustering and prediction workflows.

PROJECTS

ShadowFix: Real-World Shadow Removal | *PyTorch, OpenCV, U-Net, Gradio*

- Co-developed evaluation across traditional correction, RGB, residual, and mask-guided U-Nets on 540 held-out ISTD triplets using PSNR, SSIM, and region-specific metrics.
- Analyzed the Mask-Guided U-Net's 30.189 dB PSNR and 0.9514 SSIM, with +3.01 dB overall and +2.21 dB shadow-region PSNR over the RGB baseline.
- Tested automatic RGB-only and painted-mask Gradio workflows and documented mask-quality and out-of-distribution failure modes.

Video Anomaly Detection | *Python, PyTorch, OpenCV, CNN-LSTM*

- Built an end-to-end video pipeline combining frame sampling, background subtraction, optical flow, CNN spatial features, and LSTM temporal modeling for anomaly scoring.
- Modularized preprocessing, sequence creation, training, and inference to support reproducible experiments on recorded and live video streams.

Clinical Risk Prediction | *Scikit-learn, FastAPI, React, Docker*

- Benchmarked Random Forest, SVM, and gradient-boosted classifiers with standardized preprocessing and validation, achieving 0.92 ROC-AUC.
- Exposed the selected model through a containerized FastAPI service and built a React interface for validated, user-facing inference.

PUBLICATION

"Smart Health Prediction System with Data Mining." Co-author, *Nanotechnology Perceptions*, Vol. 20, S15, 2024. Article